AUSTIN TONSAY

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EDUCATION:

California State Polytechnic University, Pomona *Expected Graduation: May 2028*

Bachelor of Science, Aerospace Engineering (Emphasis: Astronautics) | GPA: 3.83

TECHNICAL PROJECTS:

Baseline vs Altitude-Optimized High-Power Rocketry Experiment: *May 2026 – Present*

- Designed baseline and altitude-optimized high-power rockets predicted to reach 3,095 ft and 4,523 ft (~46% increase)
- Performed aerodynamic/stability trades across fin geometry, nose profile, body length, boattailing, and surface finish in SolidWorks/OpenRocket.
- Iterated 5 custom motor-retention concepts and developed a bayonet-style retainer plus interchangeable 6-in modular avionics bay; manufactured PETG-CF aerodynamic components with DFM and anisotropic loading
- Calibrated simulations using component-level as-built mass/CG measurements and flight-tested baseline vehicle; converted vehicle loss after successful launch and apogee deployment into revised electronic tracking and high-visibility recovery requirements.

EXPERIENCE:

Design, Build, Fly (DBF) Competition: *August 2025 – Present*

Structures/Manufacturing Engineer

- Developed 3 weighted material trade studies comparing 4–6 candidates across bending/torsional strength, mass, cost, and manufacturability to guide wing/fuselage structural material selection.
- Designed tail assembly in SolidWorks intended for 3D printing to minimize part count, enhance manufacturability, and improve aerodynamic performance.
- Owned wing/fuselage assembly, wood/foam core construction with carbon-fiber reinforcement, and controlled surface finishing with Monokote heat-shrink covering.

Bronco S.T.A.R. — BLADE Odyssey: *August 2025 – May 2026*

Communications & Thermals Engineer

- Developed a heater trade study comparing 3 candidate heating solutions against a ~9.25 W / 37.2 Wh CubeSat power constraint, balancing thermal capability, power draw, size, and integration risk.
- Sized two 60 × 40 mm heating pads at 4.2 V, 3 W each (~0.71 A) to maintain thermal margin within the available system power budget.
- Designed thermostat-control pseudocode to cycle battery heating between 0–10°C with a 30%-of-flight runtime failsafe for ~4-hour CubeSat mission.

Undergraduate Missiles, Ballistics, and Rocketry Association (UMBRA): *September 2025 – November 2025*

Small Rocket Competition, Team Lead

- Designed model rocket fins and body geometry in OpenRocket, SolidWorks CAD, and Bambu Studio; improved projected apogee from 2,100 ft to 2,900 ft (~38% increase) through aerodynamic optimization.
- Led 5-person design/build team through aerodynamic design, manufacturing, and avionics integration while maintaining project schedule within 2 days of planned milestones.
- Programmed Arduino IDE for avionics systems to read rocket altitude, surrounding pressure, and temperature; post-build failure analysis identified insufficient epoxy cure time as the cause of internal structural separation.

ACADEMIC PROJECTS:

ARO1021L - Compared OpenRocket, static-fire analytical, inclinometer, and onboard altimeter apogee estimates across 3 model rockets; quantified method-to-method discrepancies and evaluated drag, manufacturing, and environmental sources of error.

EGR1000L - Designed/programmed an Arduino-controlled animatronic system integrating 4 servos and an ~85% 3D-printed structure; iterated actuator selection and mechanical fit to achieve 3 coordinated motions within a 12-in design envelope.

SKILLS:

SolidWorks, OpenRocket, FDM Additive Manufacturing, Arduino IDE (C++), PrusaSlicer, Carbon Fiber Layups, Wood/Foam Airframe Fabrication, Excel VBA